

Post-Detection report for 2021 IAA SETI Permanent Committee meeting, 27 Oct 2021, Dubai

Report given remotely by Kathryn Denning, PhD. (Associate Professor, Anthropology, York University, IAA SETI Committee member since 2005, SETI Institute Science Advisory Board member)

Prefatory note, revisiting last discussion in 2019... and some thoughts about next directions.

Currently, we do not have a post-detection working group within the IAA SETI Permanent Committee, although members continue to work independently on post-detection subjects. When our new Exec has had the opportunity to get more settled in, hopefully this subject can be revisited, with some clear objectives.

Meanwhile:

Obviously, the COVID-19 pandemic has had many effects. In addition to causing personal losses of loved family members, friends, and colleagues, and major social disruptions, it has resulted in the postponement of research meetings, delays in individual research projects, etc. Hopefully the next couple of years will be better in terms of coordinated post-detection-related work.

The last post-detection update to the SETI Permanent Committee was 2019, produced by me and Les Tennen. The view articulated (generally based on a 2019 IAC review paper, by members of this Committee¹) was that given the many changes in the news/media environment, and the changes in who is conducting SETI and how (an increase in the number and types of players, including students and citizen scientists), we should now consider a three pronged approach:

- (a) timeless Principles which hopefully don't need revision
- (b) Code of Conduct for active technosignatures researchers, which is revised periodically as circumstances require
- (c) Best Practices for Communication with the public, which are revised every couple of years

(b) and (c) could be produced by small working groups, reasonably quickly, building on existing knowledge, codes, and practices, and we could then consider how much to revise the existing Principles.

¹ Denning, Berea, Capova, Forgan, Korpela, Oman-Reagan, Tarter, and Tennen 2019. "SETI and Post-Detection: Towards a New Research Roadmap." International Astronautical Congress 2019, <https://iafastro.directory/iac/paper/id/52210/summary/>

In terms of the bigger picture: there is a great deal of basic research to be done which could help guide policy concerning post-detection matters. Therefore, Denning *et al* 2019 also suggested some research roadmapping concerning subjects like: the current post-detection concerns of scientists, students, and citizen scientists involved in technosignatures research; information seeking and sharing behaviour as practiced by scientists, the public, and governments; areas of concern for the public, e.g. what questions would the public actually ask in different detection scenarios (some of which are new), and what types of answers would be most satisfying? Hopefully some workshops on this will come together over the next couple of years.

Also in terms of the bigger picture: IAA SETI PC is not itself a research-conducting organization, and within our group, we don't always have the specific expertise to tackle particular post-detection questions ... but there is a ton of research happening in the world with direct applicability to our concerns. I think it would be helpful, therefore, for members who are motivated, to review relevant scholarship from outside astronomy and provide annual or occasional updates to the rest of the membership. This might enhance our ability to have interdisciplinary conversations and draw from relevant outside research.

During today's meeting, instead of a working group report *per se*, I will offer some (inevitably non-comprehensive) observations about current and recent (past 2 years) developments which are relevant to post-detection, and there will be two reports from members concerning their current/upcoming projects.

Just for context, here's a blast from the past: "Summary of Proposed Studies on the Implications of Peaceful Space Activities for Human Affairs", Brookings Institute, 1960: <https://ntrs.nasa.gov/citations/19620001145>

pg 42S "Though intelligent or semi-intelligent life conceivably exists elsewhere in our solar system, if intelligent extraterrestrial life is discovered in the next twenty years, it will very probably be by radio telescope from other solar systems. Evidences of its existence might also be found in artifacts left on the moon or other planets. The consequences for attitudes and values are unpredictable, but would vary profoundly in different cultures and between groups within complex societies; a crucial factor would be the nature of the communication between us and the other beings... Clearly, the better we can come to understand the factors involved in responding to such crises the better prepared we may be."

44S "While the discovery of intelligent life in other parts of the universe is not likely in the immediate future, it could nevertheless happen at any time. Whenever it does occur its consequences for earth attitudes and values may be profound. Hence a long-term research effort, which would aid in preparing for this possibility, could usefully begin with: . . . A continuing determination of emotional and intellectual understanding and attitudes regarding the possibility and consequences of discovering intelligent extraterrestrial life."

Overview of today's update

1. Papers on themes related to Post-detection in 2021 and 2020 IACs

2. Some relevant recent and upcoming activities of SETI Permanent Committee members including:

- SETI Institute and Breakthrough Listen, continued public outreach etc.
- PSETI activities
- Technosearch website
- 1I/'Oumuamua
- The Galileo Project

3. Some recent work of relevance in the larger research community

Some relevant articles and conference video.

4. Some topics of general relevance:

- a) Increasing and changing discussion around UAPs
- b) The COVID-19 pandemic: applicable lessons about mis/dis/information
- c) Scientists in society: lessons from climate and from COVID
- d) Ongoing diversification of potential detection scenarios

5. Member reports

- a) Daniela DePaulis, Stelio Montebugnoli, Andrea Melis, on A Sign In Space
- b) John Elliott on UK SETI Research Network activities related to post-detection

6. BLC1 (Breakthrough Listen Candidate 1) update: still not aliens, but informative

- * brief comments from KD about the significances of this case
- * opportunity for Q&A with Breakthrough Listen team members

1. Papers on themes related to Post-detection in 2021 and 2020 IACs

2021: <https://iafastro.directory/iac/browse/IAC-21/A4/2/> with presenter name (follow links for full author teams)

Steps to follow after establishing contact with extra-terrestrials, Mr. Aditya Balasubramaniam

The need for a worldwide and international SETI Journal, Dr. Claudio Maccone

Potential Technologies Through SETI and Global Response Plan for First Contact, Dr. Ugur Guven

"The readiness is all" - Understanding Post-Detection Dynamics through Live Action Role Play, Ms. Kate Genevieve

On the necessary aspect of METI, Dr. Carl DeVito

Astropolitics and the Implications of Belonging to an Extraterrestrial Polity, Mr. Mclee Kerolle

How law and ethics would need to adapt to the discovery of life, Mr. Jordi Sandalinas

Benedict XVI and SETI, Dr. Paolo Musso

2020: <https://iafastro.directory/iac/browse/IAC-20/A4/2/> , with presenter name (follow links for full author teams)

The Duty Under the Space Law Treaty Regime to Disclose or Report an Encounter With an Extra Terrestrial Intelligence, Mr. George Anthony Long

Protocols to be Implemented in Extraterrestrial Contact and SETI Security Protocols, Dr. Ugur Guven

The InCosmiCon Research Center and its activities in the field of SETI, Big History and interculturality, Dr. Paolo Musso

2. Some relevant recent and upcoming activities of SETI Permanent Committee members

n.b. This compilation is not comprehensive.

Also see, in sections 5 and 6 below, the separate reports on:

- c) Daniela DePaulis, Stelio Montebugnoli, Andrea Melis, on A Sign In Space
- d) John Elliott on UKSRN activities
- e) BLC1

The SETI Institute and Breakthrough Listen continue their extensive public outreach efforts.

Many of us are eagerly awaiting the results of the Astronomy and Astrophysics Decadal Survey (Astro2020) in the USA: www.nationalacademies.org/our-work/decadal-survey-on-astronomy-and-astrophysics-2020-astro2020

In 2019, members of the Committee submitted multiple papers concerning technosignatures, including one on the need for sustained post-detection research funding.

The wait is nearly over: “The decadal survey will be publicly released on Thursday, November 4, 2021 at 11am ET followed by a public briefing webinar from 2-3:30pm ET.” www.nationalacademies.org/our-work/decadal-survey-on-astronomy-and-astrophysics-2020-astro2020

To see the input that our community sent:

See <https://baas.aas.org/astro2020-apc> for

“Panoramic SETI: An all-sky fast time-domain observatory” by S. Wright et al

“The Breakthrough Listen Search for Extraterrestrial Intelligence” by Gajjar et al

“Preparing for the Discovery of Life Beyond Earth” by Denning and Dick

“The Social Sciences Interdisciplinarity for Astronomy and Astrophysics — Lessons from the History of NASA and Related Fields” by Berea et al

“Searches for Technosignatures: The State of the Profession” by J. Wright et al

and several papers on the GBT, multiple papers on exoplanets and on astrobiology

See <https://baas.aas.org/astro2020-science> for multiple science papers.

PSETI (Penn State) activities include:

PSETI 2021 symposium postponed: now June 27–30, 2022. Program TBA, but perhaps there will be a post-detection breakout session <https://sites.psu.edu/setisymposium2021/>

SETI Bibliography project: See Lafond, Wright, and Huston for the latest: <https://arxiv.org/abs/2107.02887>

SETI-News. Macy Huston and Jason Wright provide this very valuable service: <https://seti.news/about/>

SETI seminars

The Assembly of the Order of the Octopus, a student-run online meeting in July 2021

www.pseti.psu.edu/the-order-of-the-octopus-2021/ for students and early-career technosignatures

researchers. Archive at: <https://zenodo.org/communities/octopusassembly2021/search?page=1&size=20>

The meeting included several papers which touched on post-detection-related topics:

- Keynote from PC member Denise Herzing, “Animal Communication as an Analog for E.T. Signals”, <https://doi.org/10.5281/zenodo.5160121>
- Panel: Interdisciplinary SETI, with moderator Claire Webb, PC members Denise Herzing, Ravi Kopparapu, Kathryn Denning (written contribution), and Chelsea Haramia (written contribution)
- PC member Julia Demarines and collaborator Chelsea Haramia, “Speaking for Humanity” <https://doi.org/10.5281/zenodo.5156065>
- George Profitiliotis, “Private Space Activities and Solar System Technosignatures: Anticipating a Potential Wild Card Event”, <https://doi.org/10.5281/zenodo.5145138>
- Hung Lu Chan, “The Embassy of Aliens”, <https://doi.org/10.5281/zenodo.5208860>,
- Jessica Wahl, “Scientists' Understanding of Ethics and Responsibility in Astrobiology: An Empirical Study”, <https://doi.org/10.5281/zenodo.5156106>
- Rebecca Charbonneau, “Cold War Searches for Intelligence”, <https://doi.org/10.5281/zenodo.5156051>

Technosearch: <https://technosearch.seti.org/> For anyone who missed its launch in 2019: obviously a wonderful resource. In the present context: it is useful for thinking about different types of detections. How is this going, i.e. are papers being submitted? It could also be a useful model for collecting post-detection-related research.

1I/‘Oumuamua – Briefly: first known interstellar object, discovered Oct 2017, observed for a couple of months until it was out of range (including by SETI Institute on ATA and BL team on GBT), resulting in a flurry of papers 2019-2021 reporting its oddities, mostly suggesting it was probably of natural origin (e.g. Desch & Jackson, Mar 2021 <https://doi.org/10.1029/2020JE006807> , vs Avi Loeb *et al*’s theory in scientific papers, popular articles, and in Jan 2021 book *Extraterrestrial: The First Sign of Intelligent Life Beyond Earth*. Considerable press coverage has resulted, as well as some friction in the scientific community.

Strictly in terms of what this can tell us about post-detection issues – regardless of one’s favoured interpretation – the 1I/‘Oumuamua case has some relevant features. For example, it demonstrates these challenges: a phenomenon which cannot be investigated as fully as some would like, *i.e.* only temporarily accessible for study; inevitable lag time between discovery and full analysis; and a major difference of opinions between professional scientists, which has been very public. It could be a useful case study concerning the public understanding of science. It also reinforces the long-existing theoretical argument that study of physical objects in our solar system should be a focus of technosignatures research.

The Galileo Project for the Systematic Scientific Search for Evidence of Extraterrestrial Technological Artifacts, launched July 2021, led by A. Loeb, with a considerable team.

<https://projects.iq.harvard.edu/galileo>

“The goal of the Galileo Project is to bring the search for extraterrestrial technological signatures of Extraterrestrial Technological Civilizations (ETCs) from accidental or anecdotal observations and legends to the mainstream of transparent, validated and systematic scientific research. This ground-based project is complementary to traditional SETI, in that it searches for physical objects, and not electromagnetic signals, associated with extraterrestrial technological equipment.”

Active research is in early stages. Meanwhile, there are public events, popular articles, including Loeb’s recent comments concerning a “Protocol for Contact with Extraterrestrial Equipment”. Some links:

- <https://projects.iq.harvard.edu/galileo>
- www.smithsonianmag.com/science/wonder-avi-loeb-180978579/
- Upcoming online event, *Our Future in Space*, with Loeb, Bill Nelson (new NASA Administrator) and David Wilkinson <https://tix.cathedral.org/TheatreManager/1/online?performance=25007>
- <https://medium.com/on-the-trail-of-the-saucers/protocol-for-contact-with-extraterrestrial-equipment-e0ab2747ac4a>

In terms of relevance to post-detection: obviously, every new search opens up additional potential discovery scenarios. Loeb notes that “Currently, there is no international agreement on how humanity should engage with a visiting object of extraterrestrial origin. It would be prudent to formulate guidelines before they are needed.” (<https://medium.com/on-the-trail-of-the-saucers/protocol-for-contact-with-extraterrestrial-equipment-e0ab2747ac4a>)

There may be relevant sessions / papers at next spring’s SSOCIA (Social and Conceptual Issues in Astrobiology) and AbSciCon. AbSciCon will be taking paper abstracts soon.

3. Some recent work of relevance in the larger research community – not comprehensive.

n.b. Jason Wright's "SETI in 2020" overview is helpful, including the full gamut of papers.

<https://arxiv.org/pdf/2107.07512.pdf>

B. Alper and J. Alvarado. 2021. "Religious Americans less likely to believe intelligent life exists on other planets" www.pewresearch.org/fact-tank/2021/07/28/religious-americans-less-likely-to-believe-intelligent-life-exists-on-other-planets/

Amedeo Balbi, Claudio Grimaldi 2020. Quantifying the information impact of future searches for exoplanetary biosignatures. <https://arxiv.org/abs/2008.07586>

Jim Benford 2021. "A Drake Equation for Alien Artifacts." *Astrobiology* 21(6). DOI: 10.1089/ast.2020.2364

"I propose a version of the Drake equation to include searching for alien artifacts, which may be located on the Moon, Earth Trojans, and Earth co-orbital objects. The virtue of searching for artifacts is their lingering endurance in space, **long after they go dead...** Scenarios are quantified that suggest that looking for alien artifacts near Earth is a credible alternative approach relative to listening to stars."

Richard B. Bilder, (2020). "On the Search for Extraterrestrial Intelligence (SETI)". *American Journal of International Law*, 114(1), 87-95. doi:10.1017/ajil.2019.86

"However, even if the arguments against the international community's now seeking a binding international instrument regarding SETI-related issues are persuasive, there are nevertheless good reasons why the UN and COPUOS should currently consider adopting some type of nonbinding measures concerning such issues." See bio: <https://secure.law.wisc.edu/profiles/rbbilder>

Michael Bohlander "Joining the "Galactic Club": What Price Admission? – A hypothetical case study of the impact of human rights on a future accession of humanity to interstellar civilisation networks". *Futures* Volume 132, September 2021, <https://doi.org/10.1016/j.futures.2021.102801>

Michael Bohlander "Metalaw – What is it good for?" *Acta Astronautica* Volume 188, November 2021, Pages 400-404. <https://doi.org/10.1016/j.actaastro.2021.06.039>

Peter Hatfield, Leah Trueblood. 2020. SETI and Democracy. <https://arxiv.org/abs/2007.13515> Very useful article concerning replies to ETI, and public survey data.

C. Kenney and A. Lau. Pew Research Center, 2021: "Most Americans believe in intelligent life beyond Earth; few see UFOs as a major national security threat" www.pewresearch.org/fact-tank/2021/06/30/most-americans-believe-in-intelligent-life-beyond-earth-few-see-ufos-as-a-major-national-security-threat/

Brian McConnell. 2020. **The interstellar communication relay**. International Journal of Astrobiology , Volume 19 , Issue 6 , December 2020 , pp. 419 – 422 DOI: <https://doi.org/10.1017/S1473550420000178>

“The paper describes the architecture for a data repository and distribution system to be used in the case of a SETI detection event. This system is conceptually modelled after the Deep Space Network, although the hardware and infrastructure involved are different and substantially less expensive to operate. The system is designed to accommodate a large number of users from a variety of fields who wish to contribute to the analysis and comprehension effort that would follow the detection of an information-bearing signal.” (See also: www.centaury-dreams.org/2020/08/31/what-if-seti-finds-something-then-what/)

Carlos Santana, 2021 **“We Come in Peace? A Rational Approach to METI”**, *Space Policy*. Volume 57, August 2021, 101430

Kelly Smith and John W. Traphagan. 2020. **First, Do Nothing: A Meek Protocol for First Contact**. *Space Policy* 54. <https://doi.org/10.1016/j.spacepol.2020.101389>

John Traphagan and Ken Wisian. 2020. **“Protocols for Encounter with Extraterrestrials: lessons from the Covid-19 Pandemic.”** *JBIS* 73(7).
www.researchgate.net/publication/342276181_PROTOCOLS_FOR_ENCOUNTER_WITH_EXTRATERRESTRIALS_lessons_from_the_Covid-19_Pandemic See also www.centaury-dreams.org/2020/07/10/what-can-seti-scholars-learn-from-the-covid-19-pandemic/

John Traphagan 2021 **“Should We Lie to Extraterrestrials? A Critique of the Voyager Golden Records”**
<https://doi.org/10.1016/j.spacepol.2021.101440> *Space Policy* Volume 57, August 2021, 101440

Ken Wisian and John W. Traphagan. 2020. **The Search for Extraterrestrial Intelligence: A Realpolitik Consideration**. *Space Policy* 52. <https://doi.org/10.1016/j.spacepol.2020.101377> See also “On SETI, International Law, and Realpolitik” www.centaury-dreams.org/2020/05/22/on-seti-international-law-and-realpolitik/ (A response paper is likely to follow.)

Also, some seminars/conferences (of course, also see SETI Institute page for many more talks):

The Whale Sanctuary Project (Lori Marino et al) 2021. Seminar on **“Sanctuaries and the Study of Cetacean Communication”** including some remarkable initiatives: <https://whalesanctuaryproject.org/sanctuaries-and-the-study-of-cetacean-communication/>

METI International workshop: **“SETI: Cognition And Communication In Extraterrestrial Intelligence”**, 2020.
<http://meti.org/en/schedule>

Technosignatures seminar series at Goddard includes relevant talks:
<https://seec.gsfc.nasa.gov/Events/technosignatureSeminars.html>

Some papers at **SSOCIA 2020**: <https://www.ssocia.space/ssocia-2020>

4. Some Current Topics of General Relevance:

a) Increasing and changing discussion around UAPs (as discussed elsewhere in today's meeting) is an important part of the landscape. The release of this American report in June 2021 garnered much interest and much coverage. www.dni.gov/files/ODNI/documents/assessments/Preliminary-Assessment-UAP-20210625.pdf

b) The COVID-19 pandemic: applicable lessons about mis/dis/information

Unfortunately, the pandemic has required a great deal of applied research concerning the spread of misinformation and disinformation – building upon a substantial body of work that already existed concerning, in particular, the various effects of social media. Some of this will certainly be applicable to conspiracy theories concerning aliens, and potentially to mis/dis/information spread in the event of a real detection or detections. This is an excellent topic for research.

c) Scientists in society: lessons from climate and from COVID

Obviously, scientists have been playing very important roles over the past two years in doing research, communication with government, and with the public, on these two existentially crucial topics. However, at the same time, recent surveys demonstrate that an alarming number of scientists communicating with the public about divisive topics – like climate change and COVID-19) – have experienced harassment, online abuse, hostility, threats. e.g. www.nature.com/articles/d41586-021-02741-x Other research shows this phenomenon is particularly prevalent for women in the public eye... and that it is worsening. This is worth bearing in mind, since a definite detection of ETI might result in a gamut of reactions. As noted in Denning et al 2019 and Forgan and Scholtz 2016, social media safety for researchers is thus a concern.

d) Ongoing Diversification of Potential Detection Scenarios

Significant new projects and instruments mean more and different potential detection scenarios. Significant new players in SETI: in particular, first SETI observations at FAST in recent years, and the rising profile of SETI work in China. Dr Di Li gives a great update here about FAST activities including SETI, Feb 2021:

www.youtube.com/watch?v=G0cJ14HA4Xg in the IAF session on “Breakthrough Listen and International SETI Collaborations”. From a post-detection perspective: given public attention to actual and perceived rivalries between the USA and China, it's important to be able to provide reassurance that a detection of ETI at FAST would be transparent.

Obviously, the astronomers present at this meeting know the details of new technosignatures search strategies and instruments. There are other important developments too, that have everyone buzzing: e.g. the JWST, when finally launched (current estimate is Dec 2021), could quickly result in detections of biosignatures on exoplanets, which could affect thinking about the prevalence of ETI. The point here, for post-detection purposes, is that *they all result in different detection scenarios which may have significantly different characteristics and may require different public discussions.*

5 (a) A Sign in Space, by Daniela DePaulis, Stelio Montebugnoli, Andrea Melis

Full presentation at:

<https://drive.google.com/file/d/16fHGWZHImc8oik7OWXQIQ6LeR01Z8Lll/view>

5 (b) John Elliott, UK SETI Research Network Post-Detection Activities

see PPT for details.



UK SETI RESEARCH NETWORK
POST-DETECTION ACTIVITIES 2019-21

"WORKING NATIONALLY FOR INTERNATIONAL EFFECT."

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6. BLC1 (Breakthrough Listen Candidate 1, aka “Mysterious Signal from Proxima Centauri”) update: still not aliens, but informative

For the latest.... From an email to Committee members from Steve Croft, scroft@berkeley.edu 25 Oct 2021: *“We are delighted to share the results and data from our completed analysis of the “BLC1” signal detected at Parkes. Two papers in Nature Astronomy, led by UC Berkeley postdoc (and incoming SETI Institute NSF Postdoctoral Fellow) Sofia Sheikh and UCB REU student (and Hillsdale College undergrad) Shane Smith, in addition to a Research Note of the American Astronomical Society detailing some follow-up observations, are linked from <https://breakthroughinitiatives.org/news/33> and <https://seti.berkeley.edu/blc1> - where you can also find related artwork, flowcharts, videos, public data, and instructional materials.”*

Some SETI news broke in *The Guardian* in mid-Dec 2020, amid all kinds of other news -- then-record levels of second-wave COVID devastation, protests over the US federal election, the run-up to Brexit, wildfires, explosions in Afghanistan, Arecibo’s collapse, the Hayabusa2 sample return mission, Chang’e 5 lunar sample return mission. Quoting “an individual in the astronomy community who requested anonymity”, as well as some named BL members, the Guardian broke a SETI story:

Ian Sample, author: www.theguardian.com/science/2020/dec/18/scientists-looking-for-aliens-investigate-radio-beam-from-nearby-star 18 Dec 2020, **“Scientists looking for aliens investigate radio beam ‘from nearby star’”**.... *“Astronomers behind the most extensive search yet for alien life are investigating an intriguing radio wave emission that appears to have come from the direction of Proxima Centauri, the nearest star to the sun.... Analysis of the beam has been under way for some time and scientists have yet to identify a terrestrial culprit such as ground-based equipment or a passing satellite.”*

Luckily, the article was reasonable and measured, and quoted Pete Worden as saying: *“These signals are likely interference that we cannot yet fully explain. Further analysis is currently being undertaken.”*

Much media coverage rapidly ensued – most of it informed, careful, and respectful of the team’s clear message that it was likely RFI but that analysis was still underway and would take some time. e.g. *Scientific American* (Jonathan O’Callaghan and Lee Billings) posted a carefully detailed and cautious story the same day, quoting several Breakthrough members (Milner, Worden, Siemion, Sheikh), and additional astronomers Wright (Penn State) and Chatterjee (Cornell). www.scientificamerican.com/article/alien-hunters-discover-mysterious-signal-from-proxima-centauri/ “Alien Hunters Discover Mysterious Signal from Proxima Centauri”. And Nadia Drake for Nat Geo, 18 Dec: www.nationalgeographic.com/science/article/alien-hunters-detect-mysterious-radio-signal-from-nearby-star, and many more. It is a lesson in how many different ways a headline can be written...

Importantly, many astronomers – including our members – quickly posted explanatory articles on their blogs and on popular sites (Shostak, Beasley, Wright, Marchis, Garrett, and more). A Wikipedia page went up quickly...

There was some associated discussion about post-detection protocols (e.g. Leonard David’s 25 Jan 2021 interview with Worden here: www.space.com/proxima-centauri-signal-breakthrough-listen-pete-worden-interview

Sofia Sheikh provided an exemplary talk at Breakthrough Discuss in April 2021 (the conference theme was “Alpha Centauri System: A Beckoning Neighbor”), freely available online.

After that, there seems to have been a remarkable quiet until the papers were released 25 Oct 2021. The authors note: *“The blc1 signal is the first signal of interest from the Breakthrough Listen programme that required extensive signal verification to be undertaken. This case study led to a novel signal verification ‘toolkit’ for future SETI signals of interest.... Conversely, this signal of interest also reveals some novel challenges with radio SETI validation.”* www.nature.com/articles/s41550-021-01508-8 Many news articles have followed quickly.

Post-detection relevance? The Breakthrough Listen team will have additional technical comments and reflections for us. From a social science perspective, **there are some takeaways for post-detection:**

- thanks are due to many excellent journalists and science writers for their caution, accuracy, and professionalism in communicating about this signal
- kudos: a rapid response from the BL team and other astronomers was effective in getting accurate information out to the public in Dec 2020/Jan 2021
- this case reiterates the challenges for SETI PD, including premature and ill-timed leaks to media, the fact that this science takes time (i.e. prolonged analysis time, and particularly if follow-up observations seem warranted), and explaining complex technical aspects to non-experts, etc.
- this is a very instructive case for us all, perhaps particularly for those conducting SETI who do not have rapid access to media relations expertise or existing relationships with science journalists
- if the signal had passed more of the tests, or had seemed complex, there could have been more room for speculation, more for everyone to handle, and as always, this community should be working on anticipating and preparing for such situations
- Q for the BL team: the Kaggle challenge which has followed has involved many more citizen scientists, and thus may further expand potential detection scenarios?
- Q for the BL team: are people downloading the data?
- a refinement to Rio2 may be worth pursuing -- or an alternative. The supplementary info for the Sheikh et al paper, “Evaluating blc1 on the Rio Scale 2.0”, runs the Rio2 score, comes up with an appropriate zero, but notes that *“Unfortunately, the Rio Scale does not actually give us much information about signals like blc1. There is a lack of granularity, noted in the original paper, for events that score between 0 and 1 in Rio Score. We propose that a new scaling system, perhaps logarithmic, should be developed to provide appropriate context and comparison for signals at blc1’s level of interest.”*

On the latter point, I’d like to emphasize, as per several previous discussions in this community, having an interdisciplinary group work on Rio, going back to first principles, could be prudent. e.g. A logarithmic scale may make sense to technosignatures researchers but not be the best way to communicate significance to the public. Sadly, this is one of the lessons of COVID, with respect to reporting deaths: logarithmic representations, while convenient, can distort public understanding (there are multiple research papers to that effect). When reporting on something much less straightforward than a numerical count, i.e. a complicated assessment of significance of a signal detection, the problem could be further compounded.